

A novel mathematical model of ATM/p53/NF- κ B pathways points to the importance of the DDR switch-off mechanisms

ADDITIONAL FILE

Biological description

After DSBs detection ATM is auto-phosphorylated on Ser1981 and dissociates from inactive homodimer to active monomers [1]. Further auto-phosphorylation of ATM on Ser367 and Ser1893, which depends on DSBs sensor MRN, is required for complete activation of ATM [2, 3]. MRN allows ATM to be attached to the damaged sites of DNA strands. In turn, ATM monomers phosphorylate two components of the complex: Nbs1 and Mre11 [4]. Signal about DNA damage is transferred through ATM to other substrates, among them cell cycle checkpoint kinase Chk2 phosphorylated by ATM on Tyr68 [5, 6]. This phosphorylation leads to auto-phosphorylation of Chk2 on Tyr383 and Tyr387 and amplification of the signal about cell cycle arrest to conduct DNA repair [5, 7]. This phosphorylation leads to auto-phosphorylation of Chk2 on Tyr383 and Tyr387 and amplification of the signal about cell cycle arrest to conduct DNA repair [5, 7].

ATM triggers the activation of Mdm2 (E3 ubiquitin-protein ligase, double minute 2 homolog) – a natural inhibitor of p53 – probably through indirect phosphorylation of the protein kinase B (Akt) [8, 9, 10]. On the other hand, ATM acts as an activator of p53 not only directly through phosphorylation of this tumour suppressor on Ser15, but also indirectly through phosphorylation of Chk2 on Ser20 [11, 12]. Moreover, ATM influences nuclear mono-phosphorylated active form of Mdm2 by phosphorylation on Ser395, leading to inactivation of Mdm2 [13]. In turn, p53 indirectly up-regulates ATM transcription and directly inhibits Chk2 synthesis [14, 15, 16]. Chk2 plays an important role in regulation of p53-Mdm2 feedback loop by promoting degradation of Mdm2 and MdmX (Mdm family member; reviewed in [17]).

An important component of ATM/p53 signalling pathway is played by Wip1. Its main role is to inactivate phosphorylated forms of p53 (Ser15), ATM (Ser1981) and Chk2 (Tyr68), reviewed in [18, 19, 20]. Wip1 inactivates p53 also indirectly through dephosphorylation of multi-phosphorylated inactive form of Mdm2 (Ser395 phosphorylated earlier by ATM) [21, 22]. This special phosphatase is linked to ATM kinase also in another way: fully active ATM phosphorylates KSRP – KH-type splicing regulatory protein – on Ser670 and Ser132 [23, 24]. KSRP positively regulates miRNAs maturation, among them miR-16 responsible for down-regulation of Wip1 mRNA [25, 23, 26]. Another factor that links ATM with Wip1 is called CREB – cAMP response element-binding protein. When phosphorylated by active ATM (Tyr100, Ser111 and Ser121), CREB positively regulates synthesis of both ATM and Wip1 [27, 28, 29, 30, 31]. Furthermore, Wip1 transcription is up-regulated by p53 [32, 30]. Wip1 is also up-regulated by another transcription factor: NF- κ B

(reviewed in [33, 19]). The main role of Wip1 in NF- κ B pathway is inhibition of transcription of NF- κ B-dependent genes, among them the genes encoding A20, I κ B α , p53 and Wip1 proteins. In contrast, ATM activates NF- κ B pathway via the I κ B kinase complex [34, 35].

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